Efficiency vs Volume Trade-off in Boost PFC Rectifier Design for Electric Scooters

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Abstract

- PFC (Power factor correction) is essential for electric scooter onboard chargers to ensure compliance with power quality standards and reduce total harmonic distortion.
- Without PFC, non-linear loads like battery chargers cause distorted current waveforms, leading to poor power factor and inefficient energy transfer.
- Designing an efficient and compact PFC presents trade-offs- Low-frequency designs suffer from large component volumes, while high-frequency switching leads to increased switching losses and thermal issues.
- To address this design challenge, a tradeoff between efficiency and volume needs to be done.

Introduction

- The problem statement is the design of the inductor in a high-power boost must balance volume **(size)** against loss (**efficiency**). Smaller inductance (L) yields a physically smaller core but increases the ripple current and core flux swing.
- This pushes the converter into discontinuous conduction(DCM) or valley-switching mode at parts of AC cycle, raising core and copper losses.
- Conversely, a large L (low ripple) keeps the converter in continuous-conduction mode (CCM) with lower stress, but requires a bigger core/winding volume.
- Design Specifications:

1. AC input voltage = 230 V_{rms}, 50 Hz, 1 phase
2. DC output voltage = 400 V
3. Rated output power = 1.2 kW
4. Minimum output power = 0.24 kW
5. Switching frequency = 500 kHz
6. Input current THD < 5%

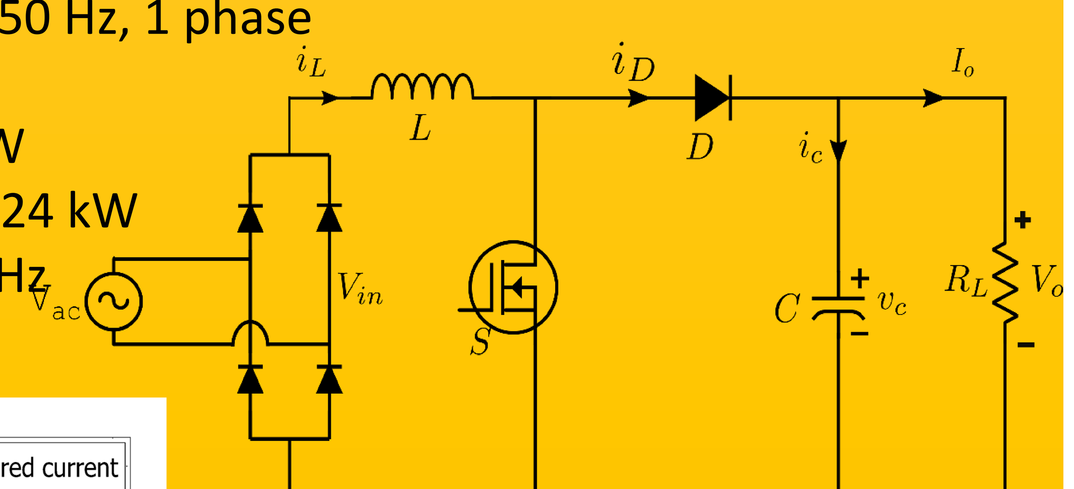


Fig1. Ideal Boost rectifier

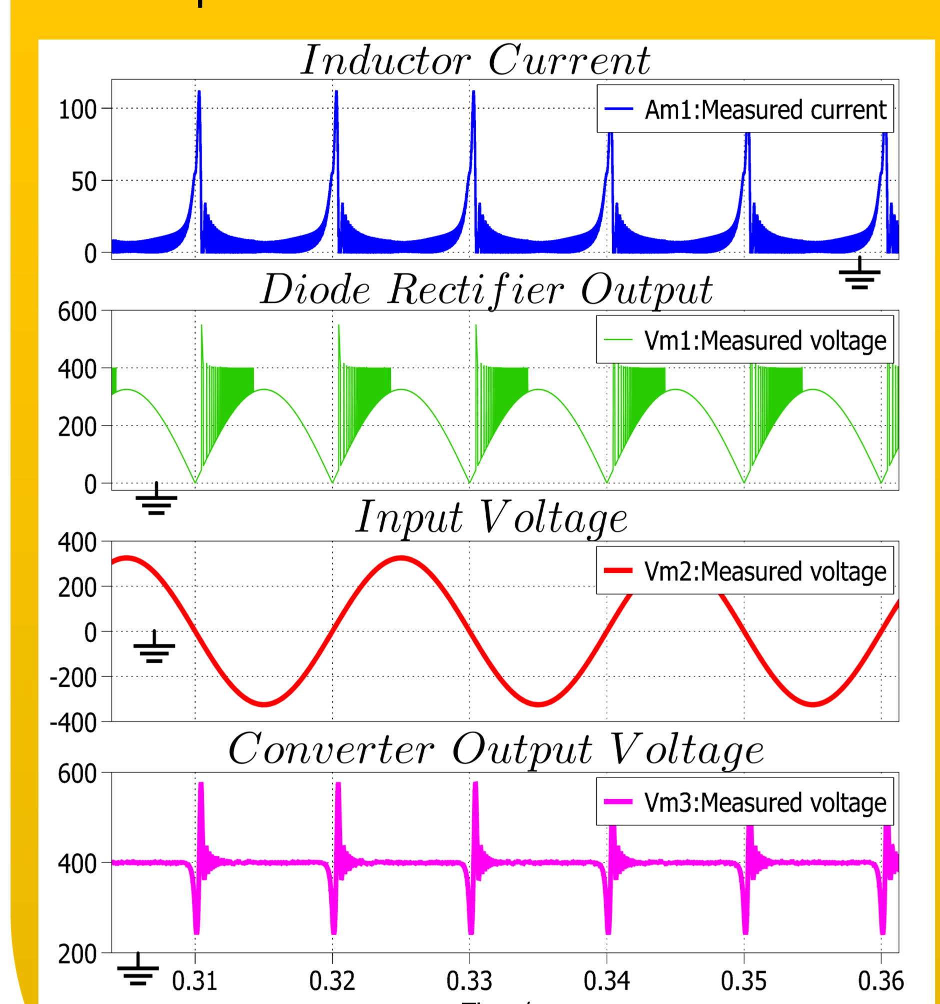


Fig2. PLECS Simulation with AC Input

Operating Point	Analysis		Simulation		Error(%)	
	ΔI_L(A)	ΔV_C(V)	ΔI_L(A)	ΔV_C(A)	ΔI_L(%)	ΔV_C(%)
Vin=10V,Po=0.24kW	1.8	0.4	1.8	0.4	0	0
Vin=10V, Po=1.2kW	1.8	2.0	1.8	2.0	0	0
Vin=325V, Po=0.24kW	11.2	0.077	11.28	0.824	0.266	90.65
Vin= 325V, Po=1.2kW	11.25	0.385	11.27	1	0.18	61.5

Table 1. Analysis vs simulation table for Vin = 10V and Po= 1200W

Methodology

- A volume based optimization methodology is adopted to minimize the physical size of the boost inductor.
- The power losses are calculated considering conduction, switching, and magnetic components to select optimal design points that lie on the Pareto volume-loss frontier.
- The switching frequency is targeted around 500 kHz which leads to lower size of passive components, but increases the switching loss.

Plots and waveforms

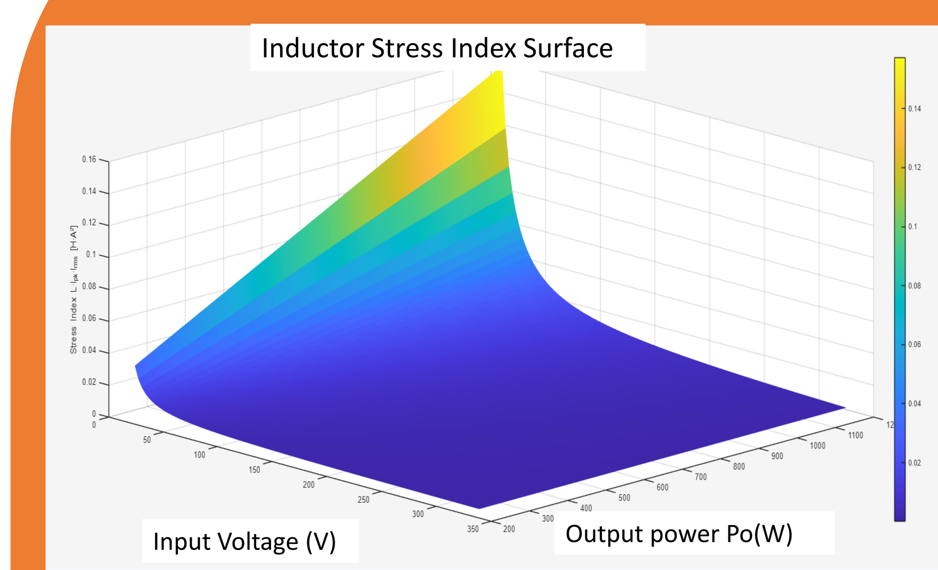


Fig.3. Plot of LI^2 vs Input Voltage and Output Power

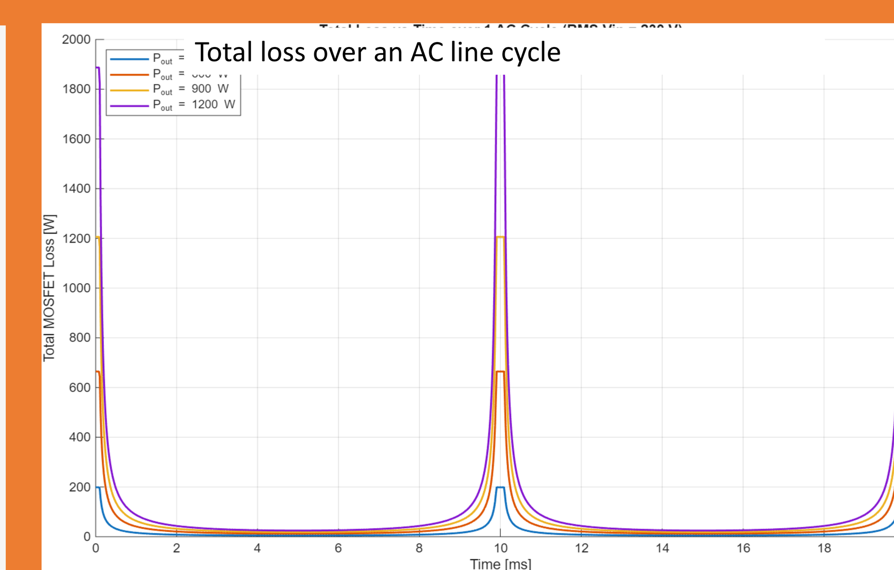


Fig.4 Total Power Loss over a line cycle

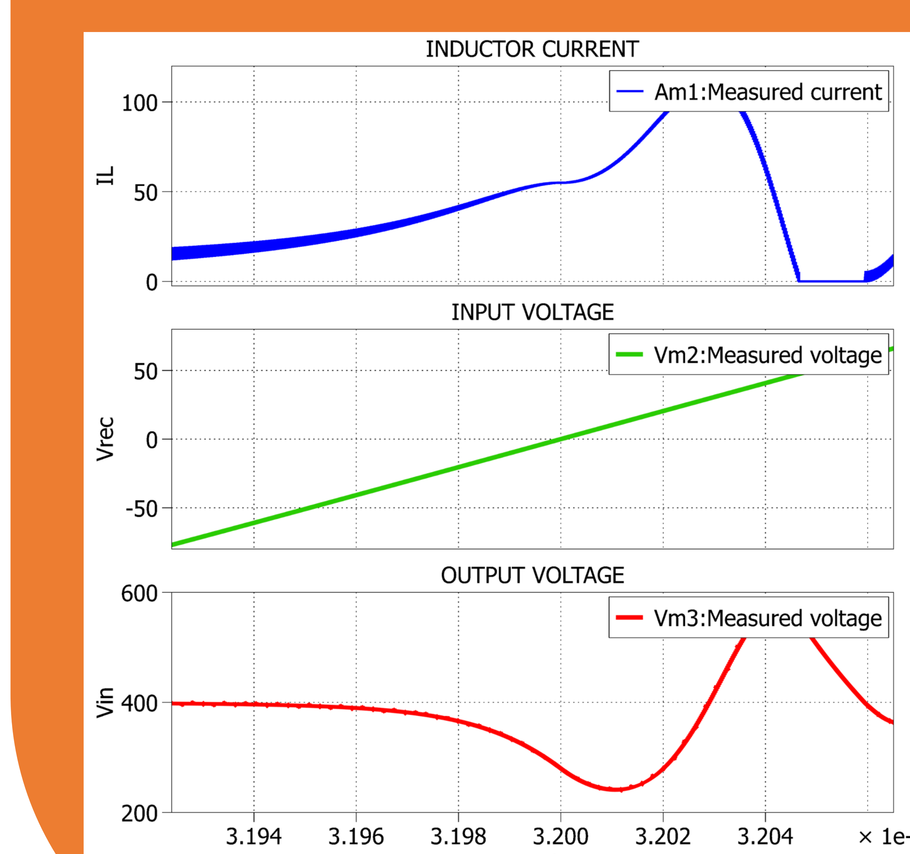


Fig.5. Inductor current and Capacitor voltage at zero crossing of AC Input

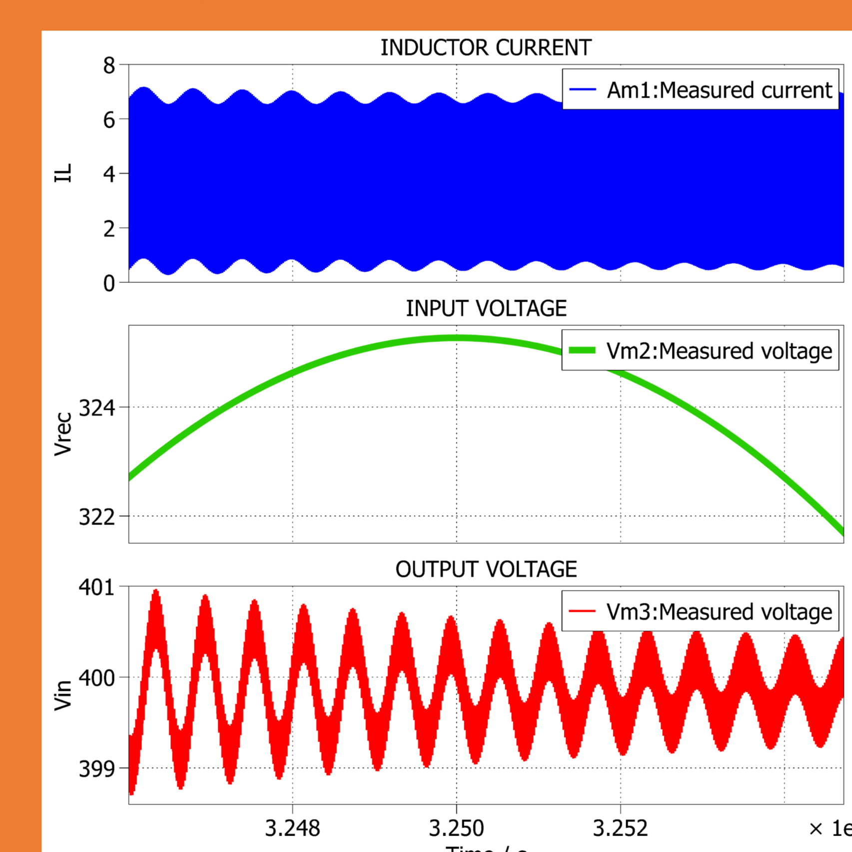


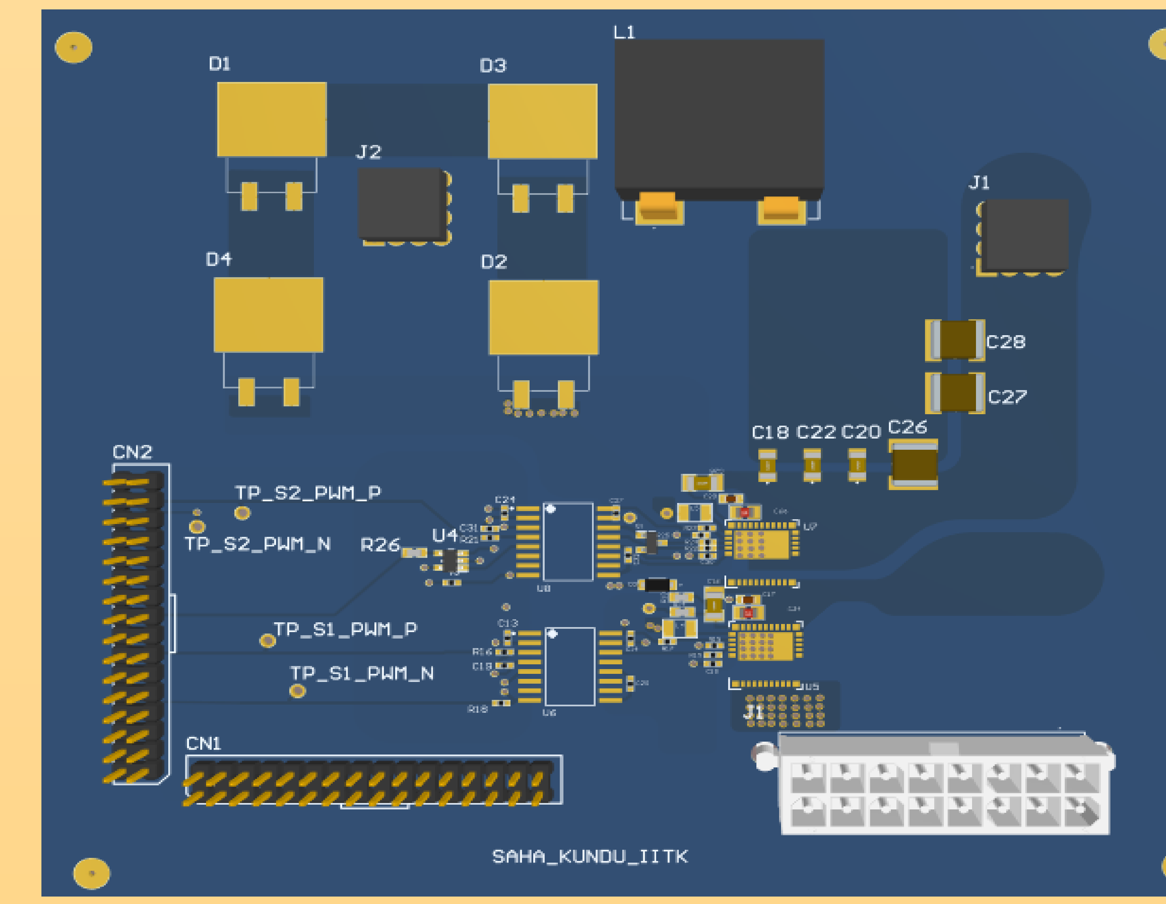
Fig.6. Inductor current and Capacitor voltage at peak of AC Input

Conclusion

- Our simulation shows that while larger inductors reduce current ripple and copper loss, they increase core size, cost, and magnetic losses.
- Conversely, smaller inductors improve compactness but at the cost of increased conduction loss and thermal stress.
- Thus, **efficiency** and **volume** must be **co-optimized** rather than maximized independently, especially in space- and thermally-constrained applications like e-mobility chargers.

Results

PCB design of the Boost rectifier is done in ALTIUM Designer.



Components	Values/Ratings	Part No.
Integrated GaNFET Switch	600V 70-mΩ	LMG3411R070
SiC Schottky Diode	650V, 10A	STPSC10H065-Y
EE40 ferrite core Inductor	20µH	B66381
HV SMD MLC Capacitor	0.01µF, 1µF	2220CC105KAZ2A, C1206C105K1RAC AUTO

Table 2. Component selection list

Future Work

Possible sources of future work:

1. Dynamic Control of Inductor Current Ripple and thermal aware optimisation
2. EMI and Acoustic noise analysis
3. ML driven Inductor design to balance conflicting metrics for optimisation.
4. Hardware prototyping, validation and system- level co-optimization

References

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